BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

SupplyLah is an autonomous, multi-agent AI orchestrator designed to digitize the informal supply chain trade within the Malaysian SME ecosystem. By leveraging Google Gemini 2.5 Flash as its central reasoning engine, SupplyLah transforms unstructured, multilingual WhatsApp communications into structured, automated workflows. The primary purpose of SupplyLah is to bridge the gap between fragmented conversational commerce and formal fintech and logistics infrastructure.

The market opportunity for SupplyLah is substantial. In Malaysia, over 1.1 million MSMEs contribute significantly to the GDP, with around 552,000 businesses in the Wholesale and Retail Trade sector alone. However, nearly 77% of these SMEs remain stuck in manual, high-friction administrative cycles. This market is driven by the rapid adoption of digital payments, the usage of WhatsApp as a primary business tool, and the national push for SME digitalization.

SupplyLah's competitive advantage lies in its Zero-Friction Agentic Architecture. Unlike traditional ERPs that require complex software migrations, SupplyLah meets merchants where they already work, on WhatsApp. By utilizing advanced Stateful Orchestration and Adaptive Negotiation logic, the system autonomously manages stock-outs, suggests substitutions, and orchestrates logistics without human intervention. This enables traditional wholesalers to scale their order volume by up to 5x while reducing human error costs by over 90%.

The business model for SupplyLah is built for scalable growth. Revenue is generated through a Tiered SaaS Subscription for wholesalers, a nominal transaction fee per successfully fulfilled order, and bespoke API connector fees for enterprise-level ERP and accounting integrations. Through this approach, SupplyLah turns the unstructured manual SME tradings into a structured, bankable digital ecosystem.

2. Product Overview / Solution

Key Features: *Highlight key features of your product*

- *WhatsApp-native order intake supporting text messages, voice notes, voice call and image-based order lists*

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- *Multilingual AI parsing for Bahasa Melayu, English, and Bahasa Rojak including slang aliases and spoken numbers*
- *Live inventory stock verification with real time cross referencing against warehouse stock levels before order is confirmed*
- *Merchant product substitution proposal engine where system proposes a pre-approved substitute product when stock is insufficient*
- Human-in-the-loop escalation with manual review modal, order correction capability and pipeline resumption after staff intervention
- Real-time Command Centre dashboard with Kanban order pipeline, exception alerts, SSE-streamed AI Reasoning Pipeline, and analytics dashboard
- AI agents that handle order parsing, stock reasoning, and logistics booking end-to-end with no manual intervention
- Automated Lalamove logistics booking with vehicle-type selection, price estimation, and real-time tracking link delivery to buyer via WhatsApp
- Serialised write queue preventing overselling under concurrent orders
- Merchant onboarding with configurable business rules (substitution mappings, discount tiers, delivery fees, and minimum order thresholds)
- Role-based staff access control with email and WhatsApp team invitation
- Dynamic few-shot semantic retrieval using pgvector (MiniLM-L12, 384-dim) grounding the Intake Agent in authentic Malaysian wholesale language patterns
- Hybrid product resolver with 3-tier matching (exact → slang alias → vector cosine similarity) with auto-resolution for single candidates
- Payment gating with simulated Touch 'n Go eWallet web payment flow before logistics dispatch
- Image-based order list parsing via Gemini 2.5 Flash Vision OCR

Technology Used: *Highlight the technology used in your application*

- *FastAPI backend containerised with Docker and deployed on AWS EC2, with Next.js frontend on Vercel*
- *Google Gemini 2.5 Flash is the central multi-agent reasoning engine and image parsing engine.*
- *Groq Whisper-v3 for real time voice transcription*
- *Supabase PostgreSQL for stateful order lifecycle management*
- *AWS S3 for media storage like audio from message intake*
- *Twilio WhatsApp Business API for buyer communication*
- *Resend for transactional staff email notifications*
- *Lalamove simulated API integration for automated delivery booking with real-time product tracking link generation*

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- *Redis for distributed inventory write queue management, ensuring overselling prevention across multiple server instances*
- *Sentence Transformers (paraphrase-multilingual-MiniLM-L12-v2) for local embedding generation and pgvector few-shot retrieval*
- *Touch 'n Go eWallet (Simulated Payment API) for payment gating before logistics dispatch*

3. Market Research & Market Dynamics

3.1 Problem Landscape

Most SME wholesalers and distributors in Malaysia use WhatsApp, sending orders via fragmented text in mixed languages. This unstructured communication creates a massive manual bottleneck for wholesalers who must interpret, cross-reference stock against physical ledgers, and manually book logistics, which is a process that takes 5 to 10 minutes per order. During peak festive seasons, this manual effort leads to mental exhaustion, inventory drift, and significant revenue loss. From a broader perspective, these SMEs lack the digital framework necessary to turn conversational commerce into structured, monetizable data, leaving them excluded from formal fintech ecosystems.

3.2 Market Niche

SupplyLah targets the traditional B2B wholesale and distribution segment in Malaysia and eventually Southeast Asia. On the supply side, our target is the mid-sized wholesaler, such as poultry, fresh produce, and dry goods, who handles 100–500 orders daily but lacks an integrated ERP. On the demand side, our target segment includes the neighborhood cafés, mom-and-pop grocery stores, and micro-merchants, who prioritize the zero-friction experience of WhatsApp over complex B2B apps. Our niche is that we occupy the space between pure messaging and complex ERP, serving those who need the intelligence of an ERP with the simplicity of a chat.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)

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TAM	Southeast Asia Countries B2B E-commerce & Wholesale Trade	RM 140.5 billion
SAM	Malaysian SME Wholesale & Retail Sector	RM 48.2 billion
SOM	Realistic market capture of tech-ready Malaysian SMEs (1.5 years)	RM 120 million

3.4 Market Dynamics & Placement:

Three key dynamics shape SupplyLah's entry and strategic placement in this market:

- **WhatsApp as the Choice of Interface:** With a 93.3% usage rate in Malaysia, WhatsApp is the go-to Operating System for local trade. SupplyLah's strategy is "Meet them where they are," ensuring zero-friction adoption. Unlike competitors who try to move users to a new app, we digitize the existing behavior.
- **Agentic AI Readiness:** Recent breakthroughs in foundational reasoning models mark a strategic transition from passive chatbots to active Agentic Orchestrators. This enables us to solve the language barrier that is mixed language, and process failures with substitution logic, all of which was technically impossible with previous-generation software.
- **Digitalization & Regulatory Push:** The Malaysian government's focus on LHDN E-invoicing and SME digitalization grants (MDEC) has created an urgent need for wholesalers to move from manual record-keeping to structured digital logs. SupplyLah acts as the agentic bridge for this transition, allowing SMEs to meet regulatory requirements and achieve operational transparency without disrupting their existing conversational workflows.

4. Value Proposition & Competitive Advantage

SupplyLah transforms wholesale order management for Malaysia SMEs by integrating directly into WhatsApp, the platform where 89.4% of Malaysian SMEs have conducted their business. Unlike traditional ERP systems that require buyers to adopt new platforms, SupplyLah requires zero behavioural change from buyers while fully automating the merchant's backend operations. The system processes orders in any Malaysian rojak languages, any unstructured format or modality sent through WhatsApp, eliminating the need for manual order-entry staff and reducing manual data entry errors by up to 10%.

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What makes SupplyLah hard to replicate is its custom built multi agent orchestration layer which is specifically built for Malaysian wholesale context. Instead of relying on generic frameworks, the system is engineered from the ground up to natively understand Bahasa Rojak, process unstructured order sent as text, voice notes or images and manage complex buyer conversations without losing context between messages. The integration between order parsing, live inventory reasoning, automated logistic booking ensures every order is handled end-to-end without any manual intervention. Also, the blend of localised AI capabilities, merchant-defined business rules, and live supply chain automation creates a pipeline that would require significant investment in both AI engineering and Malaysian market knowledge to replicate from scratch.

5. Competitor Analysis (Porter's Five Forces)

SupplyLah operates at the intersection of B2B e-commerce, traditional ERP systems and conversational AI. This analysis evaluates the competitive landscape against other players like StoreHub, Odoo, Borong and WhatsApp chatbot providers like Wati and Manychat.

Force	***Intensity***	***Assessment***
Competitive Rivalry	*High*	• *Existing manual wholesale processes are fragmented. SupplyLah differentiates by automating unstructured intake such as WhatsApp text, voice and images to significantly reduce staff manual processing time.* • *The market has heavy traditional ERPs (Odoo) and basic chatbots (Wati)* • *SupplyLah's advantage is that competitors force users to either download a new app or use rigid keyword-based bots, and our product utilises a stateful multi-agent workflow that converses in multilingual languages, automatically handles negotiations and executes logistics tool calling.*
Threat of New Entrants	*Medium*	• *While basic CRUD apps are easy to build, the ultimate barrier is the agentic workflow integration required to handle Malaysian SME wholesale complexity, unstructured data and Gemini 2.5 Flash reasoning stack.*

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		• *SupplyLah utilises atomic PostgreSQL stock deduction via a server-side RPC function that eliminates read-write race conditions at the database level. It combined with async task queues to serialise concurrent order confirmations, preventing overselling during festive demand peaks and creating a significant technical barrier to entry.*
Bargaining Power of Buyers	*Medium*	• *Wholesalers require high efficiency to maintain the margins, however, switching costs exist due to the deep integration with their inventory and established customer communication channels.* • *With SupplyLah, we offer a zero learning curve user experience, where buyers stay on WhatsApp, requiring zero onboarding for them. Besides, our system reduces human error rate of estimated at 10%, offering an immediate, measurable ROI that reduces buyer price sensitivity.*
Bargaining Power of Suppliers	*Medium*	• *SupplyLah relies on standard infrastructure (AWS, Supabase and Twilio) which are commodities, minimising dependency on a single vendor.* • *To prevent vendor lock-in and manage supplier pricing power, our backend architecture is modular, where the FastAPI orchestration layer allows us to seamlessly swap out LLM endpoints or logistic providers if supplier costs rise.*
Threat of Substitutes	*High*	• *The primary substitute is the current status quo which are the manual labour, Excel sheets and handwritten clipboards. SupplyLah must prove its ROI by reclaiming the 200+ daily orders of manual time.* • *Manual labour scales linearly, where more orders equal to more hiring staff. SupplyLah scales exponentially, so we mitigate the threat of the status quo by pricing our SaaS cheaper than hiring a single part-time data entry clerk.*

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Competitor Comparison Table:

Competitor	***Strength***	***Weakness***	***SupplyLah's Advantage***
StoreHub	Strong regional presence with built-in inventory management, business rule reasoning, and stateful order lifecycles.	Forces buyers to leave WhatsApp to use a rigid web link or app. Cannot parse unstructured "Bahasa Rojak" or handle dynamic conversational negotiations.	Brings ERP-level inventory and business rules directly to WhatsApp without requiring the buyer to adapt to new software.
Odoo	Highly comprehensive traditional ERP. Excellent for complex, structured inventory, business rules, and logistics routing.	Requires extensive staff training and forces businesses into inflexible, multi-step processes. No native WhatsApp conversational AI and cannot process messy voice notes or image-based orders.	Acts as an intelligent layer that can autonomously extract structured data from unstructured text, audio, and images to run operations.
Food Market Hub	Niche-focused on F&B and restaurant wholesale. Good structured inventory and order lifecycle management tailored for food suppliers.	Requires buyers from cafes or restaurants to log into a B2B app or portal to place orders. No WhatsApp-native AI or adaptive negotiation for stock-outs.	Allows a busy chef to simply drop a voice note on WhatsApp instead of navigating through an app catalog to place their daily order.
Borong	Dedicated B2B wholesale e-commerce marketplace. Handles structured rules, logistics, and stateful lifecycles well.	Platform-dependent. Buyers must use the Borong marketplace ecosystem. No native AI reasoning to handle	Bypasses marketplace portals by automating the supplier's existing, direct WhatsApp relationship with their buyers.

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		non-linear conversational state management.	
Wati	WhatsApp-native with basic entity extraction and simple webhooks for logistics. Great for linear customer support and broadcast marketing.	Lacks supply chain architecture. No live inventory management, no business rule reasoning, and cannot handle non-linear order lifecycles or negotiations.	SupplyLah is an agentic engine, not just a chatbot. It holds memory, checks live stock, and dynamically negotiates substitutes if an item is out of stock.
Manychat	Easy-to-use visual flow builder for WhatsApp and social media. Good basic entity extraction and simple integrations.	Purely a marketing or support tool. Fails completely if a user deviates from the scripted flow. Cannot do autonomous tool-calling or inventory reasoning.	Does not rely on rigid keyword menus. The AI understands natural intent, meaning buyers can change their minds mid-conversation and the system adapts autonomously.

6. Business Model Canvas

6.1. Customer Segmentation

1. **Wholesale Suppliers (Primary Customers)**: *SME business owners who manage over 100 daily orders manually and are bottlenecked by coordination burnout. In normal cases, owner-operated wholesale businesses in the food and grocery distribution sector in Malaysia receive 50 to 300 WhatsApp orders per day and typically 1 to 5 operation staff manage the order intake manually. Based in urban and outer-urban areas with established buyer relationships. The key pain points that are solved include lost time on manual order processing, errors during peak season and inability to scale without hiring.*

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2. **Retail Buyers (Secondary Customers):** *Restaurant operators, hawker stalls, wet market vendors and small grocery retailers who place recurring orders via WhatsApp. They are the end users of the buyer-facing interface and not the direct paying customers of SupplyLah. Their experience benefit is faster order confirmation, automatic substitution offers and no change to their existing WhatsApp ordering behaviour.*
3. **Warehouse Managers and Operations Staff (Tertiary Customers):** *Internal staff at the wholesaler who currently bear the manual processing burden. SupplyLah's dashboard and packing list notifications are designed for this user group. Their adoption drives the wholesaler's willingness to pay.*

6.2. Revenue Stream

Primary: SaaS Subscription and Usage-Based

Tier	Order Volume	Monthly Price	Target Operator
Starter	Up to 50 orders/day	RM 149/month	Small single-operator wholesaler
Growth	Up to 150 orders/day	RM 349/month	Established SME wholesaler
Pro	Up to 500 orders/day	RM 699/month	Multi-staff wholesale distributor

At 200 orders per day the AI operational cost is approximately RM 90 per month, meaning the Growth tier carries a healthy gross margin of approximately 64% on AI cost alone before infrastructure. For Pay-Per-Use transaction fee such as overages, RM 0.15 per successful order processed is applied to the client if they have used their tier limits beyond the monthly quota.

Secondary: Logistics Referral Commission

A referral commission of approximately 3 to 5% per completed Lalamove or GrabExpress booking made through SupplyLah. At 100 confirmed deliveries per day per operator at an average booking value of RM 15, this generates RM 45 or 75 per operator per day in referral revenue. However, this requires a commercial agreement with Lalamove post-MVP and is a planned Year 2 revenue stream.

Tertiary: Premium Analytics Add-On

A monthly add-on of RM 150 per operator for access to advanced reporting such as product-level margin tracking, buyer purchase pattern analysis, demand forecasting and GLM reasoning audit logs for compliance purposes. Designed for multi-branch operators and distributors who need business intelligence beyond the basic order log.

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6.3. Cost Structure

Cost Drivers	Cost Type
• Platform maintenance, Next.js dashboard, FastAPI backend, AWS EC2 and S3, Supabase infrastructure	Fixed Cost
• Google Gemini 2.5 Flash, Groq Whisper transcription, Twilio WhatsApp messaging	Variable Cost which Scales Directly with Order Volume
• AWS S3 media storage, Supabase database storage, Vercel hosting	Variable Cost which Scales with Usage
• Sales, operator onboarding support, customer success	Overhead Cost
• Product development, prompt engineering iteration, new feature development	R&D / Fixed Cost

6.4. Key Partners

Internal	External
Google Gemini 2.5 Flash	Twilio (WhatsApp Business API and Voice)
Groq (Whisper large-v3 transcription)	Lalamove / GrabExpress (logistics booking)
Supabase (database and real-time infrastructure)	Malaysian wholesale trade associations (distribution channel)
AWS (EC2 compute, S3 media storage)	
Sentence Transformers (local MiniLM-L12 embeddings)	Touch 'n Go eWallet (Simulated Payment API)

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6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Inventory Race Conditions (Overselling)	High	Medium	Implemented atomic stock deduction via a server-side PostgreSQL function that conditionally updates inventory in a single operation, eliminating read-write race conditions and preventing overselling during high-traffic bursts.
API Latency and Downtime	High	Low	Using feature toggles & Redis Queueing, where if an API fails, background workers safely buffer the payload. Fails gracefully to a "Requires Human Review" state on the dashboard.
LLM JSON Hallucination	High	Medium	Gemini 2.5 Flash occasionally returns malformed JSON, so both agents implement a 3-attempt retry loop with deep-copied message snapshots, regex trailing-comma sanitisation, and markdown fence stripping — falling back to Human Review if all attempts fail. Before any LLM call, the Intake Agent enriches the prompt with the full 57-SKU catalogue and 3 pgvector-retrieved few-shot examples (embedded via paraphrase-multilingual-MiniLM-L12-v2, CPU-only), enabling Gemini to return a structured IntakeResult with 93% confidence on Bahasa Rojak input. Extracted product names are then resolved through a 3-stage hybrid matcher — exact name, slang alias, then cosine

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			similarity — auto-proceeding at ≥80% similarity or when only one candidate exists, prompting the buyer to choose between 2+ candidates, and escalating to Human Review on no match.

7. Implementation & Financials

7.1 Go-to-Market Strategy

SupplyLah adopts a "Hub-and-Spoke" rollout strategy, targeting high-density wholesale clusters before expanding into broader regional markets. The initial launch will focus on the Klang Valley, home to Malaysia's largest wholesale markets.

The primary acquisition strategy involves Industry Vertical Pilots, partnering with specific trade associations such as Poultry or Fresh Produce Associations to onboard their core members, as well as direct operator outreach through wet market associations and wholesale trade networks where the pain of manual WhatsApp order processing is most visible and detrimental. To accelerate penetration, we will offer a Digital Clerk Trial Program, providing 60 days of free automated order processing for the first 50 wholesalers in each hub, followed by a tiered subscription model as they see measurable reductions in administrative overhead. Next, the third is buyer-side virality, as buyers continue to use WhatsApp and receive faster and more professional, structured responses, thus reinforcing the operators satisfaction with the product without any additional marketing spent.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Soft Launch	Pilot in Klang Valley Wholesale Hubs	Month 1-12	50 Wholesalers for the Digital Clerk Trial Program → 30 paying active clients
Expansion Phase	Rollout to Penang (Northern) & Johor (Southern)	13-24 Months	Expand to 200 Wholesalers across the region and convert 120 paying active clients

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Optimization	Advanced Agentic Analytics & SaaS Self-Onboarding	Month 25-36	Reach break-even per region
TOTAL	RM 10.5M investment	36 Months	~25% Market share of targeted SME Wholesalers

7.3 Revenue Projections

Metric	**Year 1**	**Year 2**	**Year 3**
Active Operator Clients	30 (10 Starter + 20 Growth)	120 (28 Starter + 80 Growth + 12 Pro)	350 (28 Starter + 260 Growth + 62 Pro)
Average Revenue Per Operator/Month	RM 282	RM 337	RM 395
SaaS Subscription Revenue	RM 101,640	RM 485,760	RM 1,659,000
Logistics Referral Commission	RM 0	RM 86,400 (roughly RM 60 per month per operator earned, with 80 orders per month)	RM 378,000 (roughly RM 90 per month per operator earned, with 120 orders per month)
Analytics Add-On Revenue	RM 0	RM 54,000	RM 210,000
Total Revenue	**RM 101,640**	**RM 626,160**	**RM 2,247,000**
Infrastructure & AI Cost	RM (32,400)	RM (108,000)	RM (252,000)
Go-to-Market Strategy Cost	RM (67,600)	RM (260,000)	RM (610,000)
Net Position	***RM +1,640***	***RM +258,160***	***RM +1,385,000***

Year 1 is intentionally loss-making where the priority is validating the product with 30 real operator clients and collecting enough order history data to enable the analytics and forecasting features that help to drive Year 2 and 3 growth.

Infrastructure & AI Cost Estimation

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External APIs and Dependencies	Cost and Analysis
Google Gemini 2.5 Flash, Groq OpenAI Whisper Large V3	For text inputs, using Gemini, the typical WhatsApp text can span from 30 to 100 words, which can account for an average of 100 input tokens, and with $0.30 per 1M input tokens and $2.50 per 1M output tokens, the cost will be $0.00003 and $0.000625, totalling to $0.000655. For audio, using the Groq OpenAI Whisper API, the cost per second is $0.00003 ($0.111 per hour), and with an estimated 30 second voice note and a 2 minute voice call, the average length is 75s, the cost per order is $0.0023. Using Gemini 2.5 Flash, the transcript text for structuring the order is about 150 tokens leading to $0.000045 ($0.30 per 1M input tokens), then outputting 250 tokens of generated JSON, costing another $0.000625 ($2.50 per 1M output tokens), bring the cost to $0.003. For images, using Gemini 2.5 Flash, the handwritten or printed list consumes about 2000 tokens, costing $0.0006 ($0.3 per 1M input tokens), then producing 250 tokens costs another $0.000625, totalling the cost to $0.0012. **Summary for Intake Handling Agent:** One order costs about $0.0016 Next, using Gemini for complex reasoning for the inventory and business logic agent, would take approximately 1200 tokens (1000 input, 200 output), costing $0.0008 without optimization. By using context caching, we can reuse the wholesaler system prompt, pricing and inventory schemas, takes about 800 tokens, and through cache hitting, the cost is billed at $0.03 per 1M input tokens. Only 200 tokens from the specific buyer's order (200 tokens for output) will be billed at the standard input rate, bringing inference costs to $0.0006 per session. **Summary for Inventory Handling and Business Logic Agent:** One order costs about $0.0006. Then, using Gemini, processing the logistics API response and giving the confirmation message to the buyer can consume 500 tokens (300 input and 200 output), the price is at $0.0006.

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	Summary for Logistics Agent: One order costs about $0.0006. The entire session of the order costs about $0.0028, where processing about 500 orders per month is about RM 5.56 (1 USD = 3.97 MYR), which is about RM 6.
Twilio WhatsApp API	At $0.005 per outbound message fee, one order can use 3 outbound messages through a combination of a debounce buffer, where 500 estimated orders per month, 1500 messages, which leads to $7.50 (RM 29.78), which is RM 30.
Cloud Service Overhead	RM 10 This involves the shared costs of the database and hosting divided across the users.
Buffer / Overhead	RM 44

Go-to-Market Strategy Cost

Metric	**Year 1: Klang Valley**	**Year 2: Regional Partners**	**Year 3: Southeast Asia**
Direct Sales & Onboarding	RM 30,000	RM 120,000	RM 240,000
Marketing & Trade Associations	RM 20,000	RM 60,000	RM 180,000
Administrative & Regional Legalities	RM 10,000	RM 30,000	RM 80,000
R&D (AI Tuning & Feature Development)	RM 7,600	RM 50,000	RM 110,000
Total Operating Expenses	***RM 67,600***	***RM 260,000***	***RM 610,000***

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Target Operator Growth	30 clients	120 clients	350 clients
Customer Acquisition Cost	RM 2,253 per operator	RM 2,166 per operator	RM 1,742 per operator

7.4 Key Financial Milestones

- Month 4: Successful completion of Klang Valley pilot with 150 active wholesalers and a 95% retention rate.
- Month 12: Expansion into Northern and Southern corridors, achieving a monthly recurring revenue (MRR) of RM 450,000 through tiered subscriptions.
- Month 18: Reaching operational break-even; AI agents successfully processing over 10,000 transactions daily across three major states.
- Month 30: Achieving positive cash flow; feasibility review for cross-border expansion into the Singapore-Malaysia-Indonesia logistics corridor.

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